



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

EXPERIMENT 6

Aim: To perform analysis of Android apps using android lint.

Theory:

The Android Lint command is a tool provided by the Android SDK that helps developers identify and correct issues in their Android projects. It performs static analysis of the code, looking for potential bugs, performance issues, usability problems, accessibility issues, and more. Lint checks can range from simple syntax errors to more complex issues that might not be immediately obvious, such as deprecated API usage, missing translations, or potential security vulnerabilities. The Android Lint tool is integrated into the Android Studio IDE, but it can also be run from the command line. This makes it accessible to developers who prefer working outside of an IDE or who need to integrate lint checks into continuous integration (CI) pipelines. When you run the Android Lint command, it scans your project's source code and resources, generating a report that lists any issues it finds. The report can be viewed in Android Studio, or it can be exported to various formats, such as HTML or XML, for further analysis or documentation.

Code:

Execute `.\gradlew lint` in the terminal of the Android Studio

```
PS C:\Users\djsce.student\AndroidStudioProjects> .\gradlew lint

Welcome to Gradle 8.2!

Here are the highlights of this release:
- Kotlin DSL: new reference documentation, assignment syntax by default
- Kotlin DSL is now the default with Gradle init
> Task :app:lintReportDebug
Wrote HTML report to file:///C:/Users/djsce.student/AndroidStudioProjects/app/build/reports/lint-results-debug.html

BUILD SUCCESSFUL in 48s
23 actionable tasks: 12 executed, 11 up-to-date
PS C:\Users\djsce.student\AndroidStudioProjects> 
```

Open Lint Report in Browser



SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



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Lint Report: 34 warnings

Check performed at Mon Apr 21 11:16:26 IST 2025 by AGP (8.2.1)

Overview

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1 ManifestOrder : Incorrect order of elements in manifest
1 RedundantLabel : Redundant label on activity
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Included Additional Checks (36)

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DISMISS

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Affected by scoped storage

[./src/main/AndroidManifest.xml:28](#): READ_EXTERNAL_STORAGE is deprecated (and is not granted) when targeting Android 13+. If you need to query or interact with MediaStore or media files on the shared storage, you should instead use one or more new storage permissions: READ_MEDIA_IMAGES, READ_MEDIA_VIDEO or READ_MEDIA_AUDIO.

```
25     </activity>
26   </application>
27   <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
28 </manifest>
```

ScopedStorage Correctness Warning Priority 8/10

EXPLAIN DISMISS

Use FragmentContainerView instead of the <fragment> tag

[./src/main/res/layout/content_main.xml:8](#): Replace the <fragment> tag with FragmentContainerView

```
5     android:layout_height="match_parent"
6     app:layout_behavior="getString/appbar_scrolling_view_behavior">
7
8     <fragment
9       android:id="@+id/new_host_fragment_content_main"
10      android:name="androidx.navigation.fragment.NavHostFragment"
11      android:layout_width="wrap_content"
12      android:layout_height="wrap_content">
13
14     </fragment>
```

Vendor: Android Open Source Project
Identifier: androidx.fragment
Feedback: <https://issuetracker.google.com/issues/new?component=600964>

FragmentTagUsage Correctness Warning Priority 5/10

EXPLAIN DISMISS

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EXPLAIN DISMISS

Incorrect order of elements in manifest

[./src/main/AndroidManifest.xml:28](#): <uses-permission> tag appears after <application> tag

```
25     </activity>
26   </application>
27   <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
28 </manifest>
```

ManifestOrder Correctness Warning Priority 5/10

EXPLAIN DISMISS

Redundant label on activity

[./src/main/AndroidManifest.xml:18](#): Redundant label can be removed

```
15     <activity
16       android:name=".MainActivity"
17       android:exported="true"
18       android:label="@string/app_name"
19       android:theme="@style/Theme.AppCompat">
20       <intent-filter>
21         <action android:name="android.intent.action.MAIN" />
```

RedundantLabel Correctness Warning Priority 5/10

EXPLAIN DISMISS



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Obsolete Gradle Dependency

[././build.gradle.kts:39:](#) A newer version of androidx.appcompat.appcompat than 1.6.1 is available: 1.7.0

```
36
37 dependencies {
38
39     implementation("androidx.appcompat:appcompat:1.6.1")
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout:constraintlayout:2.1.4")
42     implementation("androidx.navigation:navigation-fragment:2.7.6")
43 }
```

[././build.gradle.kts:40:](#) A newer version of com.google.android.material:material than 1.11.0 is available: 1.12.0

```
37 dependencies {
38
39     implementation("androidx.appcompat:appcompat:1.6.1")
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout:constraintlayout:2.1.4")
42     implementation("androidx.navigation:navigation-fragment:2.7.6")
43     implementation("androidx.navigation:navigation-ui:2.7.6")
44 }
```

[././build.gradle.kts:41:](#) A newer version of androidx.constraintlayout.constraintlayout than 2.1.4 is available: 2.2.1

```
38
39     implementation("androidx.appcompat:appcompat:1.6.1")
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout:constraintlayout:2.1.4")
42     implementation("androidx.navigation:navigation-fragment:2.7.6")
43     implementation("androidx.navigation:navigation-ui:2.7.6")
44     testImplementation("junit:junit:4.13.2")
45 }
```

[././build.gradle.kts:42:](#) A newer version of androidx.navigation:navigation-fragment than 2.7.6 is available: 2.8.9

```
39     implementation("androidx.appcompat:appcompat:1.6.1")
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout:constraintlayout:2.1.4")
42     implementation("androidx.navigation:navigation-fragment:2.7.6")
43     implementation("androidx.navigation:navigation-ui:2.7.6")
44     testImplementation("junit:junit:4.13.2")
45     androidTestImplementation("androidx.test.ext:junit:1.1.5")
46 }
```

[././build.gradle.kts:43:](#) A newer version of androidx.navigation:navigation-ui than 2.7.6 is available: 2.8.9

```
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout:constraintlayout:2.1.4")
42     implementation("androidx.navigation:navigation-fragment:2.7.6")
43     implementation("androidx.navigation:navigation-ui:2.7.6")
44     testImplementation("junit:junit:4.13.2")
45     androidTestImplementation("androidx.test.ext:junit:1.1.5")
46     androidTestImplementation("androidx.test.espresso:espresso-core:3.5.1")
47 }
```

[././build.gradle.kts:45:](#) A newer version of androidx.test.ext:junit than 1.1.5 is available: 1.2.1

```
42     implementation("androidx.navigation:navigation-fragment:2.7.6")
43     implementation("androidx.navigation:navigation-ui:2.7.6")
44     testImplementation("junit:junit:4.13.2")
45     androidTestImplementation("androidx.test.ext:junit:1.1.5")
46     androidTestImplementation("androidx.test.espresso:espresso-core:3.5.1")
47 }
```



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Obsolete SDK_INT Version Check

[././src/main/res/values-v23](#): This folder configuration (v23) is unnecessary; minSdkVersion is 24. Merge all the resources in this folder into values.

ObsoleteSdkInt

Performance

Warning

Priority 6/10

EXPLAIN

DISMISS

Unused resources

[././src/main/res/values/colors.xml](#):3: The resource R.color.black appears to be unused

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <resources>
3   <color name="black">#FF000000</color>
4   <color name="white">#FFFFFFFF</color>
5 </resources>
```

[././src/main/res/values/colors.xml](#):4: The resource R.color.white appears to be unused

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <resources>
3   <color name="black">#FF000000</color>
4   <color name="white">#FFFFFFFF</color>
5 </resources>
```

[././src/main/res/layout/content_main.xml](#):2: The resource R.layout.content_main appears to be unused

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
3   xmlns:app="http://schemas.android.com/apk/res-auto"
4   android:layout_width="match_parent"
5   android:layout_height="match_parent">
```

[././src/main/res/values/dimens.xml](#):2: The resource R.dimen.fab_margin appears to be unused

```
1 <resources>
2   <dimen name="fab_margin">16dp</dimen>
3 </resources>
```

[././src/main/res/menu/menu_main.xml](#):1: The resource R.menu.menu_main appears to be unused

```
1 <menu xmlns:android="http://schemas.android.com/apk/res/android"
2   xmlns:app="http://schemas.android.com/apk/res-auto"
3   xmlns:tools="http://schemas.android.com/tools"
4   tools:context="com.example.myapplication.MainActivity">
```

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EXPLAIN

DISMISS

Missing inputType

[././src/main/res/layout/activity_main.xml](#):11: This text field does not specify an inputType

```
8   tools:context=".MainActivity">
9   <!-- Edit text to enter course name -->
10  <EditText
11    android:id="@+id/editCourseName"
12    android:layout_width="match_parent"
13    android:layout_height="wrap_content">
```

[././src/main/res/layout/activity_main.xml](#):20: This text field does not specify an inputType

```
17    android:layout_width="match_parent"
18    android:layout_height="wrap_content">
19    <!-- Edit text to enter course duration -->
20    <EditText
21      android:id="@+id/editCourseDuration"
22      android:layout_width="match_parent"
23      android:layout_height="wrap_content">
```

[././src/main/res/layout/activity_main.xml](#):29: This text field does not specify an inputType

```
26    android:layout_width="match_parent"
27    android:layout_height="wrap_content">
28    <!-- Edit text to display course tracks -->
29    <EditText
30      android:id="@+id/editCourseTracks"
31      android:layout_width="match_parent"
32      android:layout_height="wrap_content">
```

[././src/main/res/layout/activity_main.xml](#):38: This text field does not specify an inputType

```
35    android:layout_width="match_parent"
36    android:layout_height="wrap_content">
37    <!-- Edit text for course description -->
38    <EditText
39      android:id="@+id/editCourseDescription"
40      android:layout_width="match_parent"
41      android:layout_height="wrap_content">
```

TextFields

Usability

Warning

Priority 5/10

EXPLAIN

DISMISS



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Use Autofill

[./src/main/res/layout/activity_main.xml:11: Missing autofillHints attribute](#)

```
8      tools:context=".MainActivity">
9
10     <!--Edit text to enter course name-->
11     <EditText
12         android:id="@+id/idEdtCourseName"
13         android:layout_width="match_parent"
14         android:layout_height="wrap_content"
```

[./src/main/res/layout/activity_main.xml:20: Missing autofillHints attribute](#)

```
17         android:minHeight="48dp" />
18
19     <!--edit text to enter course duration-->
20     <EditText
21         android:id="@+id/idEdtCourseDuration"
22         android:layout_width="match_parent"
23         android:layout_height="wrap_content"
```

[./src/main/res/layout/activity_main.xml:29: Missing autofillHints attribute](#)

```
26         android:minHeight="48dp" />
27
28     <!--edit text to display course tracks-->
29     <EditText
30         android:id="@+id/idEdtCourseTracks"
31         android:layout_width="match_parent"
32         android:layout_height="wrap_content"
```

[./src/main/res/layout/activity_main.xml:38: Missing autofillHints attribute](#)

```
35         android:minHeight="48dp" />
36
37     <!--edit text for course description-->
38     <EditText
39         android:id="@+id/idEdtCourseDescription"
40         android:layout_width="match_parent"
41         android:layout_height="wrap_content"
```

Autofill

Usability

Warning

Priority 3/10

EXPLAIN

DISMISS



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Hardcoded text

./src/main/res/layout/activity_main.xml:16: Hardcoded string "Enter course Name", should use @string resource

```
13         android:layout_width="match_parent"
14         android:layout_height="wrap_content"
15         android:layout_margin="10dp"
16         android:hint="Enter course Name"
17         android:minHeight="48dp" />
18
19     <!--edit text to enter course duration-->
```

./src/main/res/layout/activity_main.xml:25: Hardcoded string "Enter Course Duration", should use @string resource

```
22         android:layout_width="match_parent"
23         android:layout_height="wrap_content"
24         android:layout_margin="10dp"
25         android:hint="Enter Course Duration"
26         android:minHeight="48dp" />
27
28     <!--edit text to display course tracks-->
```

./src/main/res/layout/activity_main.xml:34: Hardcoded string "Enter Course Tracks", should use @string resource

```
31         android:layout_width="match_parent"
32         android:layout_height="wrap_content"
33         android:layout_margin="10dp"
34         android:hint="Enter Course Tracks"
35         android:minHeight="48dp" />
36
37     <!--edit text for course description-->
```

./src/main/res/layout/activity_main.xml:43: Hardcoded string "Enter Course Description", should use @string resource

```
40         android:layout_width="match_parent"
41         android:layout_height="wrap_content"
42         android:layout_margin="10dp"
43         android:hint="Enter Course Description"
44         android:minHeight="48dp" />
45
46     <!--button for adding new course-->
```

./src/main/res/layout/activity_main.xml:52: Hardcoded string "Add Course", should use @string resource

```
49         android:layout_width="match_parent"
50         android:layout_height="wrap_content"
51         android:layout_margin="10dp"
52         android:text="Add Course"
53         android:textAllCaps="false" />
54
55 </LinearLayout>
```

HardcodedText

Internationalization

Warning

Priority 5/10

EXPLAIN

DISMISS



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Included Additional Checks

This card lists all the extra checks run by lint, provided from libraries, build configuration and extra flags. This is included to help you verify whether a particular check is included in analysis when configuring builds. (Note that the list does not include the hundreds of built-in checks into lint, only additional ones.)

[LIST ISSUES](#)

[DISMISS](#)

Disabled Checks

One or more issues were not run by lint, either because the check is not enabled by default, or because it was disabled with a command line flag or via one or more `lint.xml` configuration files in the project directories.

[LIST MISSING ISSUES](#)

[DISMISS](#)



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Suppressing Warnings and Errors

Lint errors can be suppressed in a variety of ways:

1. With a `@SuppressWarnings` annotation in the Java code
2. With a `tools:ignore` attribute in the XML file
3. With a `//noinspection` comment in the source code
4. With ignore flags specified in the `build.gradle` file, as explained below
5. With a `lint.xml` configuration file in the project
6. With a `lint.xml` configuration file passed to lint via the `--config` flag
7. With the `-ignore` flag passed to lint.

To suppress a lint warning with an annotation, add a `@SuppressWarnings("id")` annotation on the class, method or variable declaration closest to the warning instance you want to disable. The id can be one or more issue id's, such as `"UnusedResources"` or `{"UnusedResources", "UnusedIds"}`, or it can be `"all"` to suppress all lint warnings in the given scope.

To suppress a lint warning with a comment, add a `//noinspection id` comment on the line before the statement with the error.

To suppress a lint warning in an XML file, add a `tools:ignore="id"` attribute on the element containing the error, or one of its surrounding elements. You also need to define the namespace for the tools prefix on the root element in your document, next to the `xmlns:android` declaration:

```
xmlns:tools="http://schemas.android.com/tools"
```

To suppress a lint warning in a `build.gradle` file, add a section like this:

```
android {  
    lintOptions {  
        disable 'TypographyFractions', 'TypographyQuotes'  
    }  
}
```

Here we specify a comma separated list of issue id's after the `disable` command. You can also use `warning` or `error` instead of `disable` to change the severity of issues.

To suppress lint warnings with a configuration XML file, create a file named `lint.xml` and place it at the root directory of the module in which it applies.

The format of the `lint.xml` file is something like the following:

```
<?xml version="1.0" encoding="UTF-8"?>  
<lint>  
    <!-- Ignore everything in the test source set -->  
    <issue id="all">  
        <ignore path="*/test/*" />  
    </issue>  
  
    <!-- Disable this given check in this project -->  
    <issue id="IconMissingDensityFolder" severity="ignore" />  
</lint>
```

Conclusion: Thus we successfully performed analysis of Android apps using android lint.